



WINDOWS SERVER LAB

Could I start you off with a beverage?...



MARCH 18, 2026
DAVID COGBILL

WINDOWS SERVER LAB

TABLE OF CONTENTS

Instructions.....	2
Create the VM	2
Configure the Server	2
Add key roles	2
Creating User account:	2
Creating Network Shares	3
Create a GPO	3
Confirm Everything is Working:	3
Wrap-up	3
Network Diagram	4
Initial Server Setup	4
Adding & Configuring DHCP & DNS Services	9
Creating Organizational Units and Groups	13
Adding Users.....	14
Adding Users to Groups	15
Creating Network Shares	17
Create GPOs	18

INSTRUCTIONS

You are tasked with creating a virtual server for a new client. This client will also need you to create accounts for all its users. They will need department shares and some basic security setup. Complete the requested build and document the entire process. Make sure to create screen shots as you work to have thorough documentation.

Create the VM

Create a VM with the following specs:

- 2 CPU Core
- 4 GB of RAM
- 50 GB HDD

Configure the Server

The server must have a static IP of 192.168.56.2, configured with a gateway of 192.168.56.1, and have been named ABCServer01. The domain should be named ABC.local

Add key roles

The server must have the following roles setup and running correctly

- AD - DS
- DNS
- DHCP

Creating User account:

Create accounts for the following users.

John Williams

Director of Operations

Human Resources

jwilliams@abc.com

David Bautista

IT Manager

Information Technology

Dave@abc.com

Samantha Rockwell

Director of Marketing

Marketing

srockwell@abc.com

Lupita Nyong'o

President

Leadership

Lupita@abc.com

Willard Smith

Director of Engineering

Engineering

wsmith@abc.com

Steven Ballmery

Marketing Intern
Marketing
sballmery@abc.com

Stephen Goss

System Admin
Information Technology
sgoss@abc.com

Jackie McMullen

Engineer
Engineering
jmcmullen@abc.com

Franz Kafka

Project Manager
Engineering
fkafka@abc.com

Janet Villanueva

Executive Assistant
Human Resources
jvillanueva@abc.com

Creating Security Groups

Each department should have a security group with each member of the department in it. Create a additional group for leadership. In it should be the President and the Department Directors. Also create a Culture Committee distribution group and populate the group with one member from each department.

All Present and future IT Department Members should have Domain Level Administrator rights. Please provide this access and a solution so that future members of this department will automatically get this access.

Creating Network Shares

Create a Network share and create a folder for each department. Make sure to set the permissions so that only department members can access their folders.

Create a GPO

Also create a GPO that doesn't allow non-IT Personnel to access Control Panel. As well as blocks users from changing wallpaper.

Confirm Everything is Working:

Create a second Windows 10 VM and join the VM to your domain and test logging in with one of your user accounts. make sure they can access the folder you created in the VM.

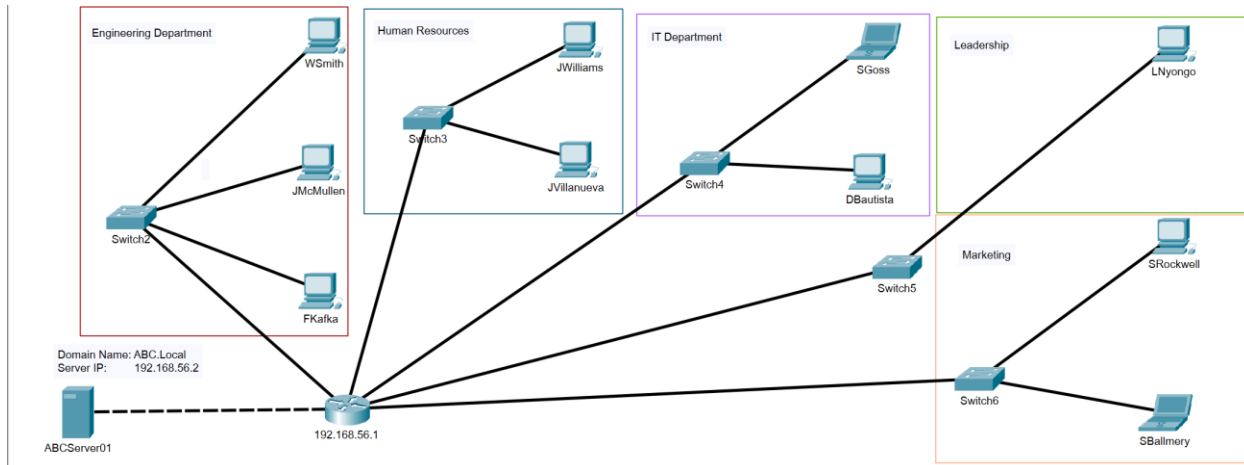
Wrap-up

Expected Documentation should be done in Microsoft Word and include:

WINDOWS SERVER LAB

- All steps taken to complete objectives. Screenshots that highlight the process. (You don't need a screenshot for every step. They should be sprinkled in like chocolate chips in a cookie.)
- Server Name, IP Address, DHCP Scope, Gateway Address, etc.
- Documentation of Administrator credentials for the server.
- Create a Network Diagram with basic information (Device type, IP information, Network Subnet, etc.) on each device. Include a fictional Router, Switch, and your virtual server.

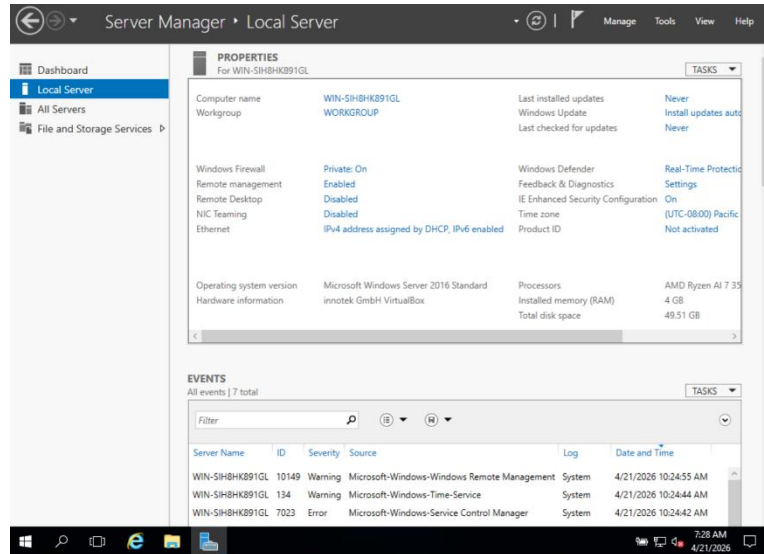
NETWORK DIAGRAM



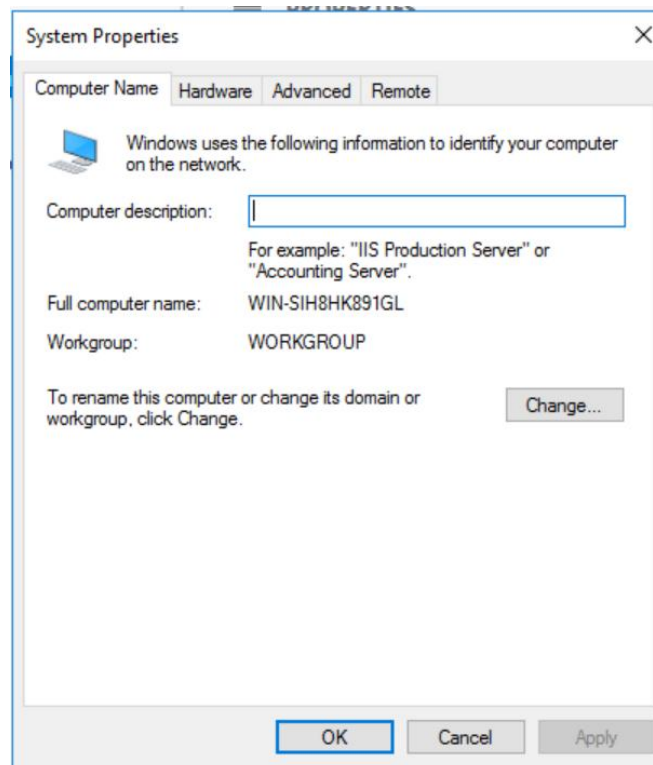
INITIAL SERVER SETUP

1. Select language: English(US), select time and currency: English(us), select Keyboard input: us, then click Next
2. Click Install now.
3. Select Windows server 2016 standard Desktop experience, Click Next
4. Check Accept license terms, then click Next
5. Select Custom install, then click Next
6. Where do you want to install windows, then click next. The computer will restart.
7. Create Administrator password, click finish, press ctr + alt + del, then enter password

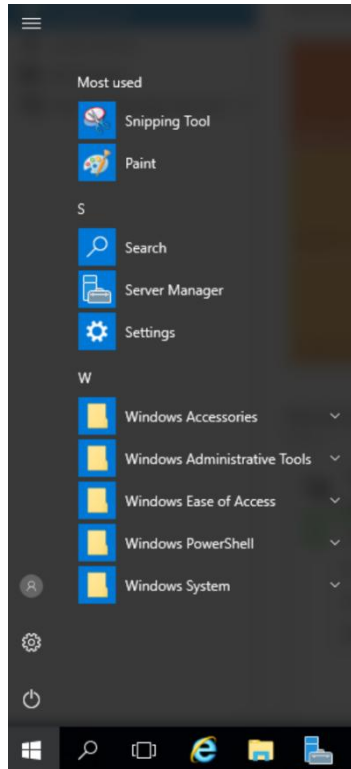
WINDOWS SERVER LAB



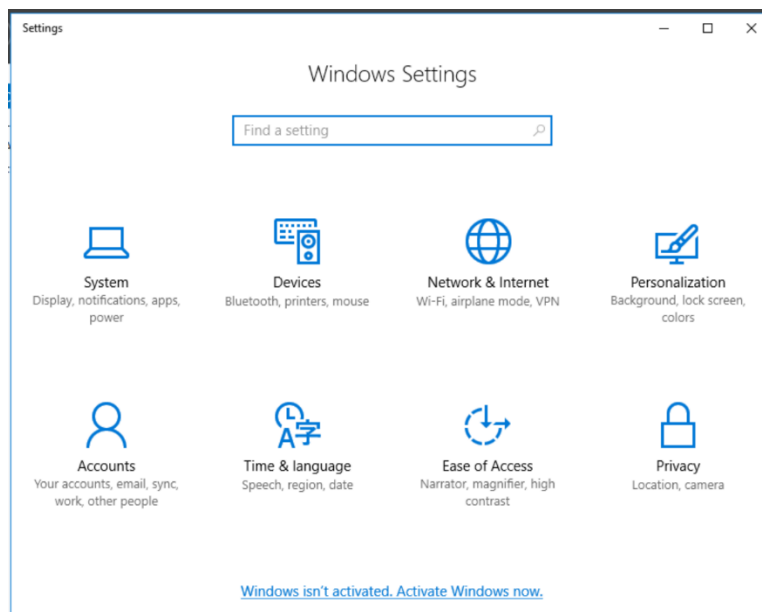
- From the Server Manager Dashboard window, click local server on the menu to the left, then click computer name



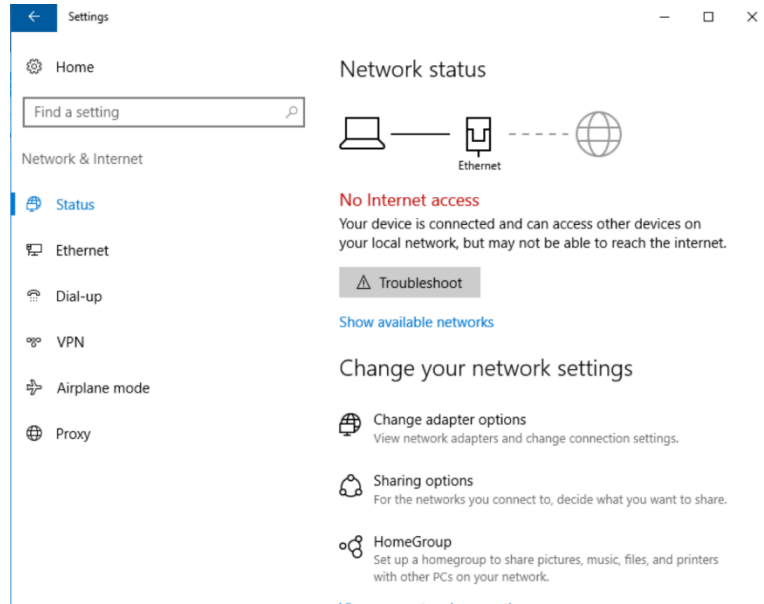
- Click change, enter name: ABCServer01, click ok, click close, Click Restart Now



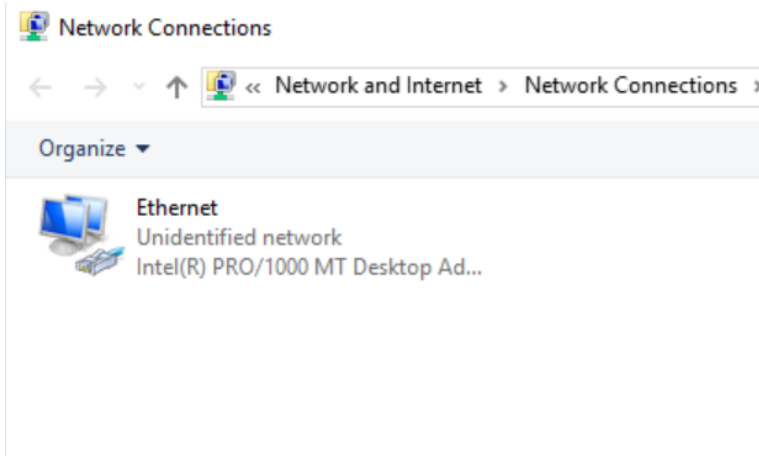
10. Click the Windows icon in the bottom left corner, click Settings(gear icon).



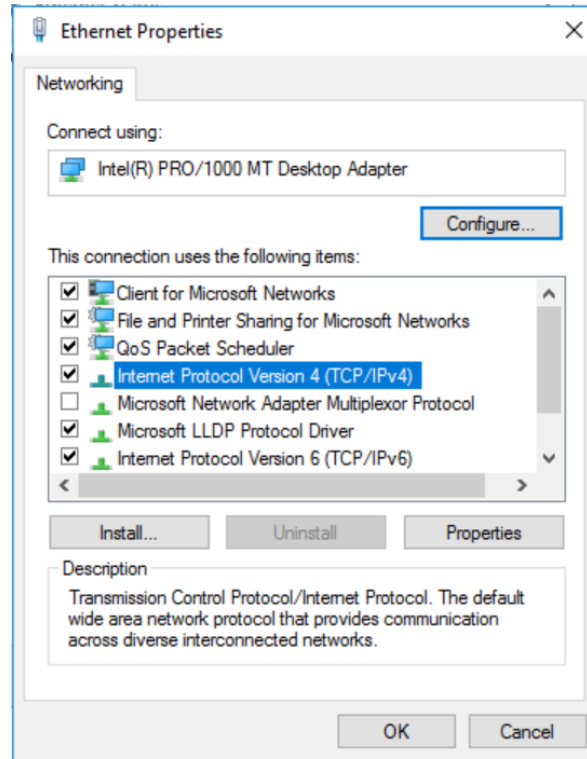
11. Click network & internet



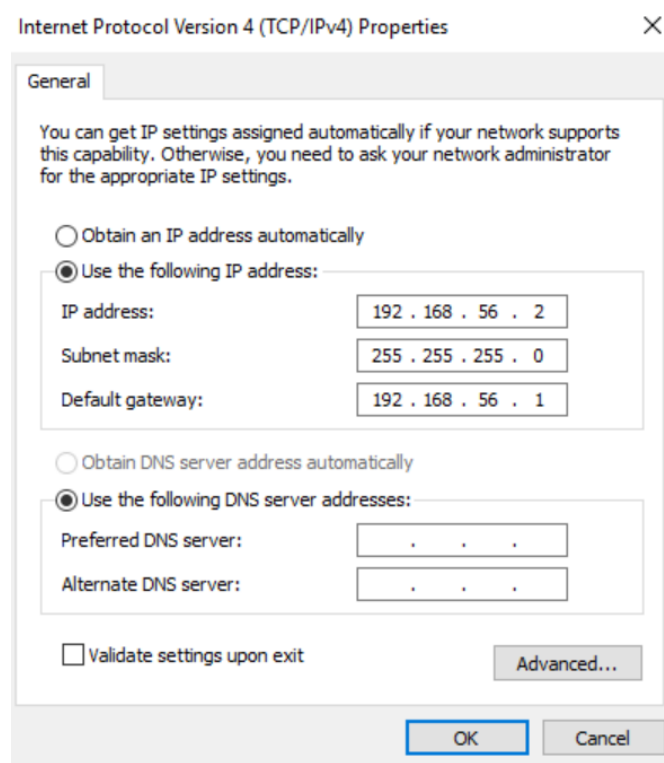
12. Click change adapter options



13. Right click ethernet, then click properties.



14. Select Internet Protocol Version 4 (TCP/IPv4

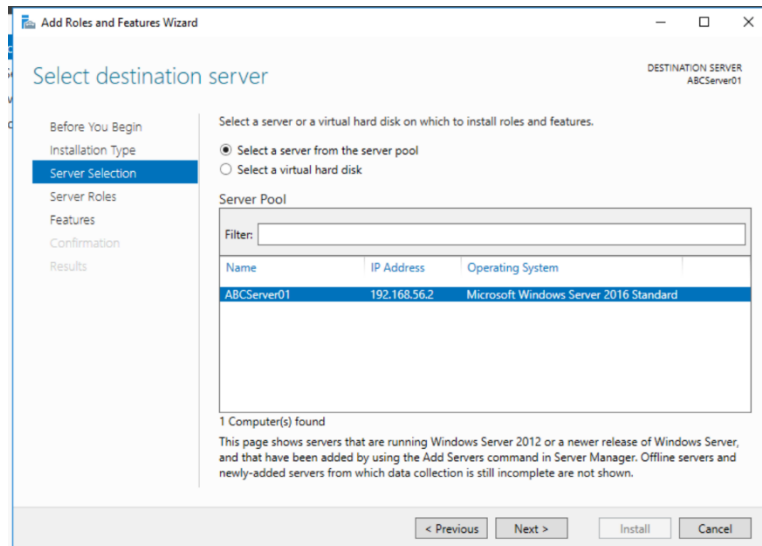


WINDOWS SERVER LAB

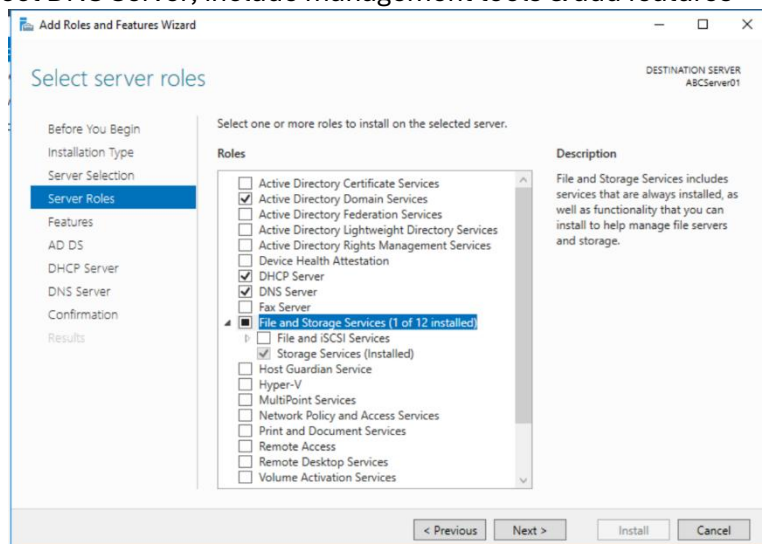
- Click properties, Use the following IP address, enter ip address & default gateway, click okay, click close

ADDING & CONFIGURING DHCP & DNS SERVICES

- From the Server Manager Dashboard window, click Add roles and features, check “skip this page by default”, Select Role based installation, then click next.

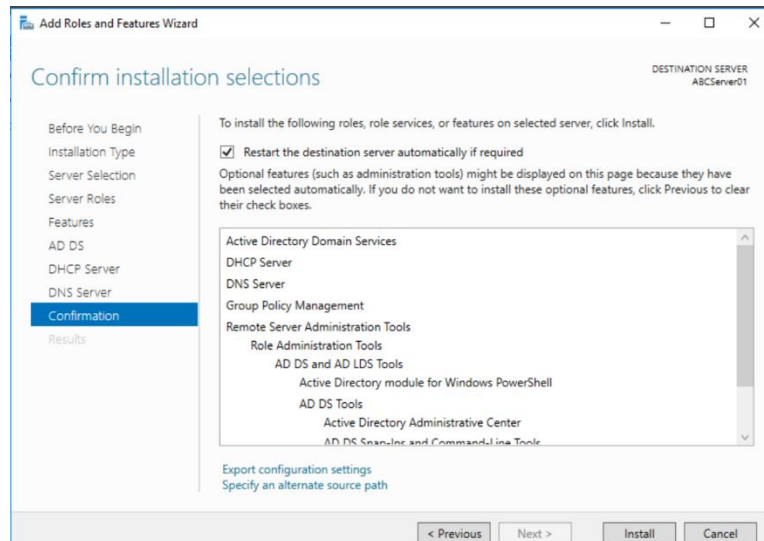


- Select server from server pool, then click next
- After selecting each service a window will appear to confirm your selection and include management tools & add features
 - Select active directory domain services, include management tools & add features
 - Select DHCP Server, include management tools & add features
 - Select DNS Server, include management tools & add features

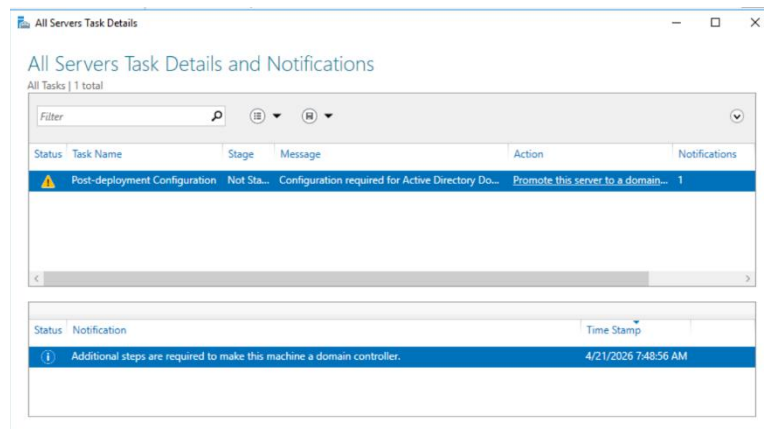


WINDOWS SERVER LAB

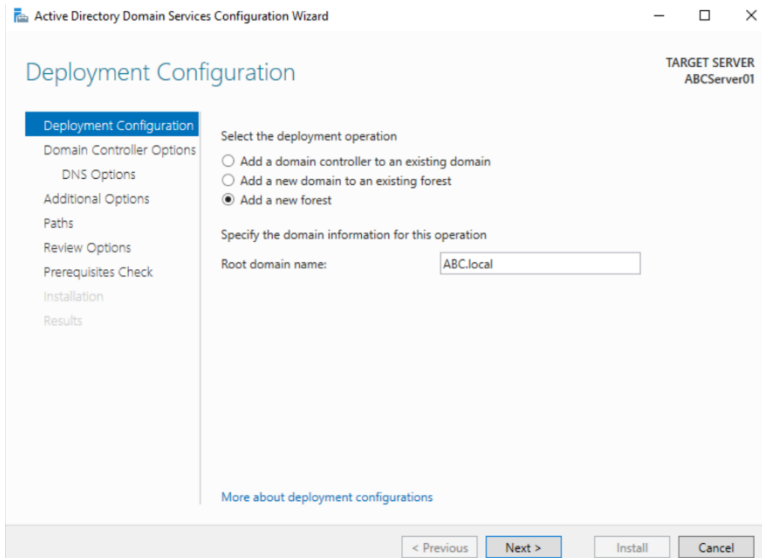
19. Click next, select features window, next, Active Directory Domain Services, click next, DHCP Server, click next, DNS Server, then Click next



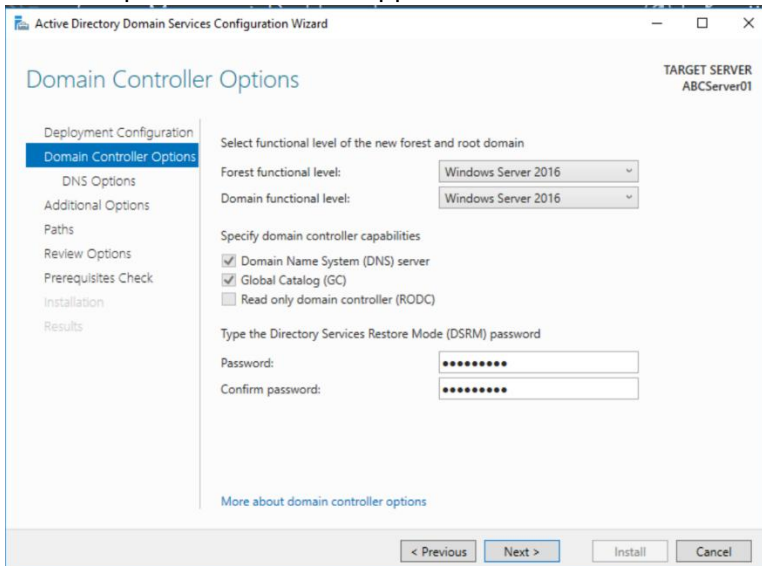
20. Click install. The Results window will appear.



21. Click Promote this server to a domain controller. The Deployment configuration window will appear.

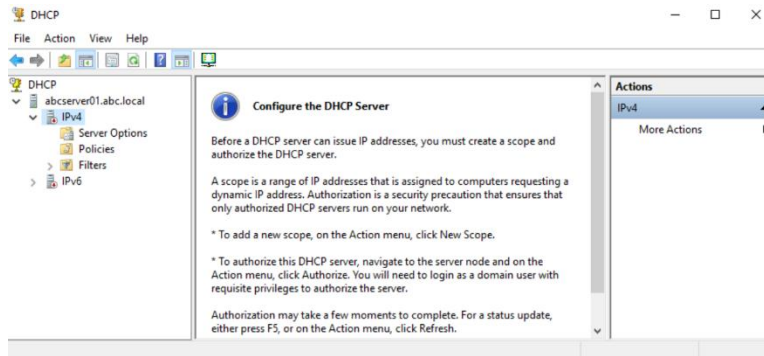


22. Select add a new forest, enter the domain name and add “.local”, then click next. The Domain Controller Options window will appear.



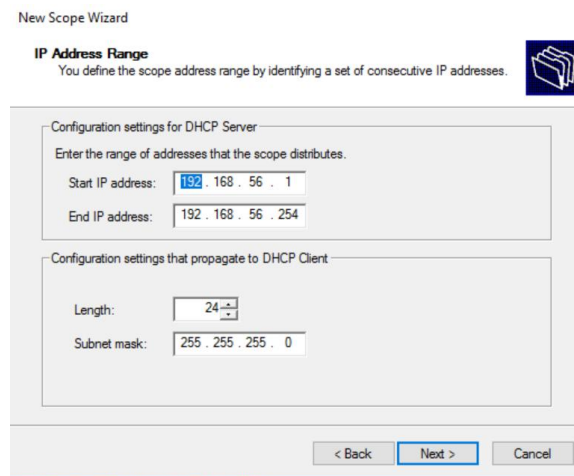
23. Enter DSRM password, next, DNS Options Window, click Next, Additional Options> Click Next, Paths, Click Next, Review Options, Click Next, Prerequisites Check, wait, Click Install, wait, close, restart
24. Use the Tools dropdown in the Server Manager window to navigate to DHCP, dropdown “abcserver01.abc.local”

WINDOWS SERVER LAB

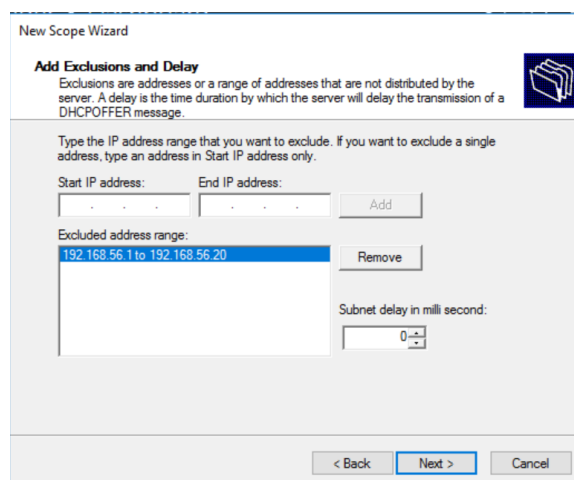


25. Right click IPv4, click New Scope, The “Welcome to New Scope Wizard” window will appear

26. Click Next, Enter Scope Name, Click Next, IP Address Range

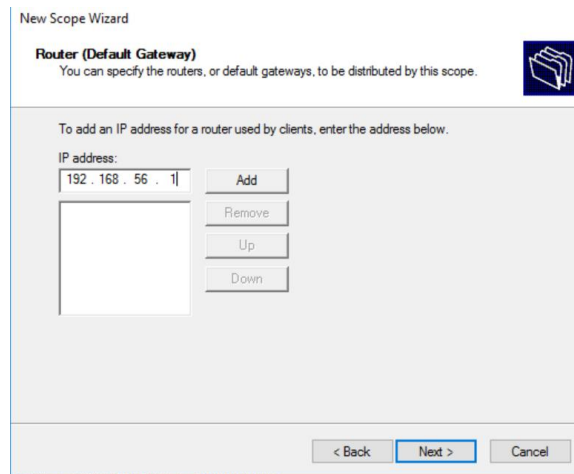


27. Enter Start IP “192.168.56.1” & end IP “192.168.56.254”, then click next.



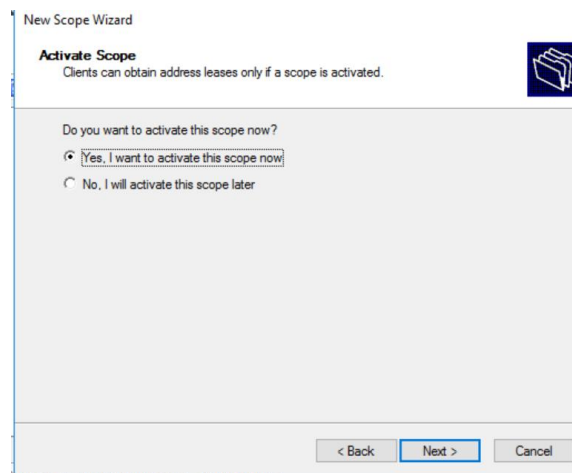
28. Enter Start Ip “192.168.56.1” and End IP “192.168.56.20”, click Add, click Next, then click Next.

29. Select “Yes, I want to configure these options now”, then Click Next.



The screenshot shows the 'New Scope Wizard' window, specifically the 'Router (Default Gateway)' step. The title bar says 'New Scope Wizard'. Below the title, it says 'Router (Default Gateway)' and 'You can specify the routers, or default gateways, to be distributed by this scope.' There is a small icon of a folder with a plus sign. Below this, it says 'To add an IP address for a router used by clients, enter the address below.' There is a text box labeled 'IP address:' containing '192.168.56.1'. To the right of the text box are four buttons: 'Add', 'Remove', 'Up', and 'Down'. At the bottom of the window are three buttons: '< Back', 'Next >', and 'Cancel'.

30. Enter “192.168.56.1”, Click Add, Click Next, Domain Name and DNS Servers, Click Next, WINS Servers, Click Next, Activate Scope –



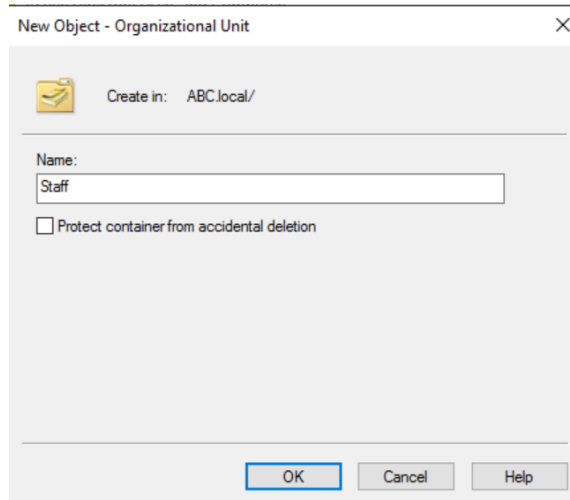
The screenshot shows the 'New Scope Wizard' window, specifically the 'Activate Scope' step. The title bar says 'New Scope Wizard'. Below the title, it says 'Activate Scope' and 'Clients can obtain address leases only if a scope is activated.' There is a small icon of a folder with a plus sign. Below this, it says 'Do you want to activate this scope now?'. There are two radio buttons: the first is selected and labeled 'Yes, I want to activate this scope now', and the second is labeled 'No, I will activate this scope later'. At the bottom of the window are three buttons: '< Back', 'Next >', and 'Cancel'.

31. Select “Yes I want to activate this scope now”, Click Next, Click Finish

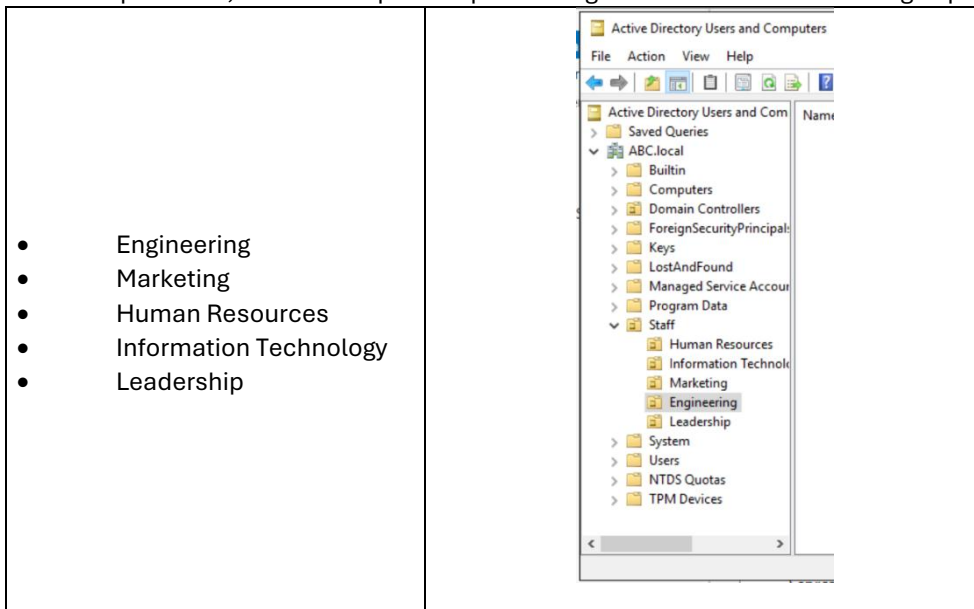
CREATING ORGANIZATIONAL UNITS AND GROUPS

1. Use the Tools dropdown in the Server Manager window to navigate to Active Directory Users and Computers.
2. Right click “ABC.Local”, hover over New, click Organizational Unit. A New Object - Organizational Unit window will appear
3. Enter Staff, Click OK

WINDOWS SERVER LAB



- Right click Staff, hover over New, click Organizational Unit
- Enter a department, click OK. Repeat steps 3 through 5 for each of the following departments:



- Use the Tools dropdown in the Server Manager window to navigate to Active Directory Users and Computers.
- Click Staff dropdown, right click a department, hover over New, click Group
- Enter department name, click OK, repeat step 7 for each department
- Use the Tools dropdown in the Server Manager window to navigate to Active Directory Users and Computers
- Right click Staff, hover over New, click Group, enter "Cultural Committee" - Group Type = Distribution, click OK

ADDING USERS

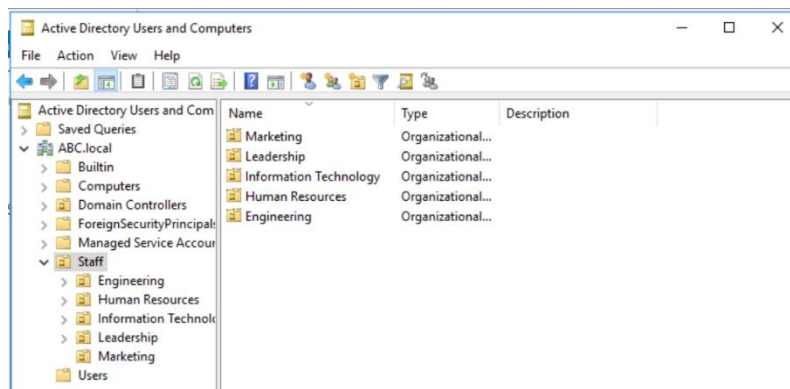
Use the following information to add users to the domain:

Name	Username	JobTitle	Department	Email
Willard Smith	WSmith	Director of Engineering	Engineering	wsmith@abc.com

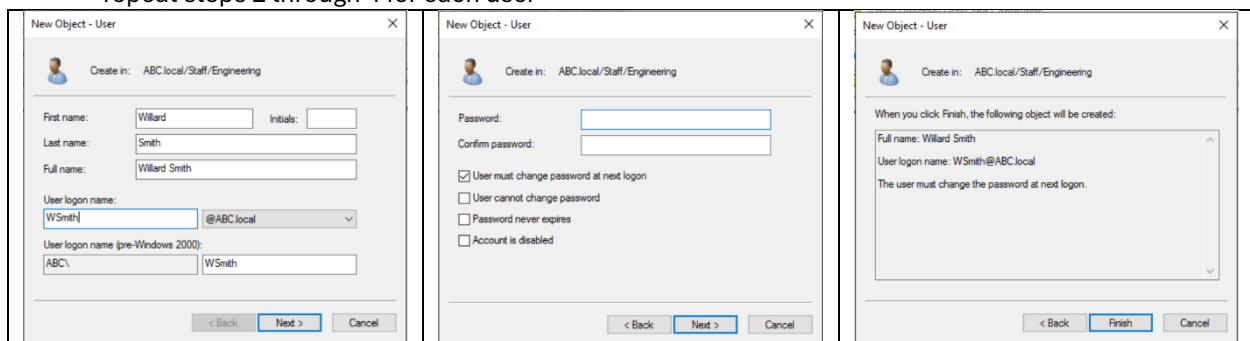
WINDOWS SERVER LAB

Samantha Rockwell	SRockwell	Director of Marketing	Marketing	srockwell@abc.com
John Williams	JWilliams	Director of Operations	Human Resources	jwilliams@abc.com
Jackie McMullen	JMcMullen	Engineer	Engineering	jmcullen@abc.com
Janet Villanueva	JVillanueva	Executive Assistant	Human Resources	jvillanueva@abc.com
David Bautista	DBautista	IT Manager	Information Technology	Dave@abc.com
Steven Ballmery	SBallmery	Marketing Intern	Marketing	sballmery@abc.com
Lupita Nyong'o	LNyongo	President	Leadership	Lupita@abc.com
Franz Kafka	FKafka	Project Manager	Engineering	fkafka@abc.com
Stephen Goss	SGoss	System Admin	Information Technology	sgoss@abc.com

1. Use the Tools dropdown in the Server Manager window to navigate to Active Directory Users and Computers.



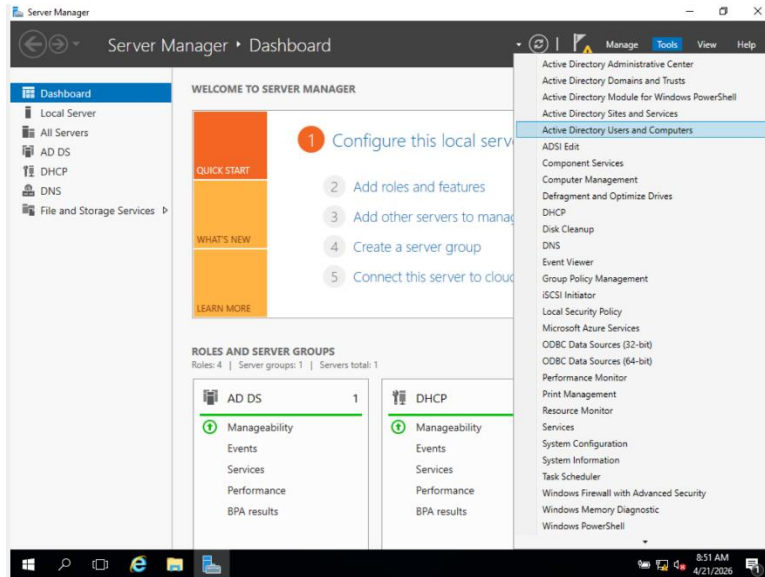
2. Click Staff dropdown, right click department, hover over New, then click User. A “New Object – User” window will appear.
3. Enter user information, click next
4. Create & enter password, Click Next. A window to confirm the user info will appear. Click Finish, repeat steps 2 through 4 for each user



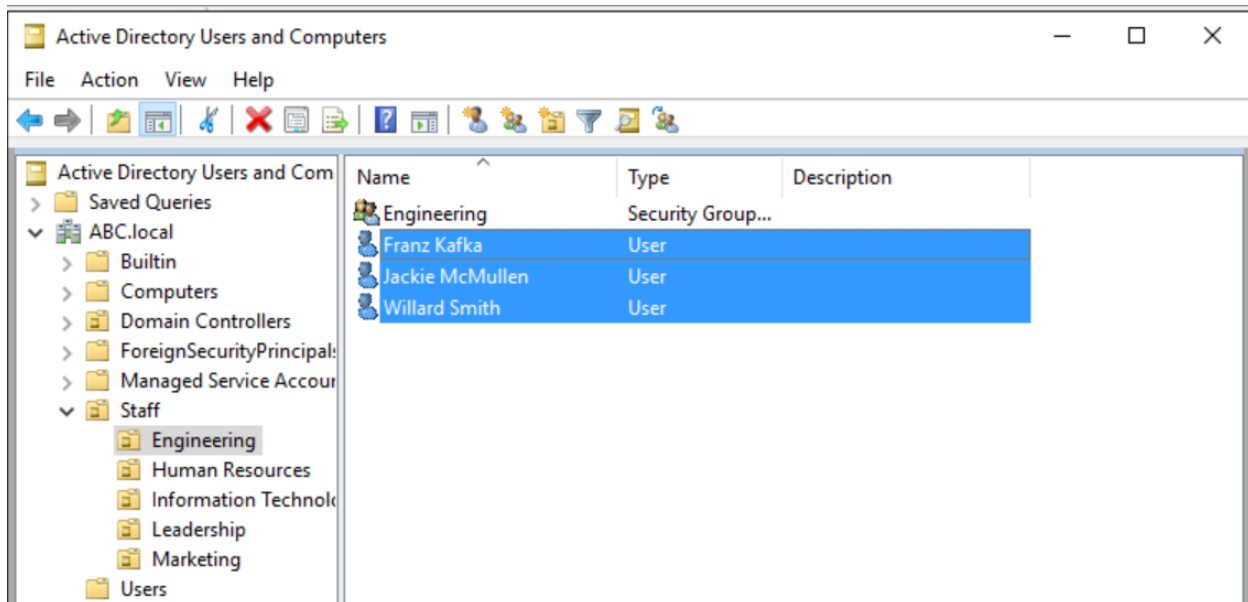
ADDING USERS TO GROUPS

1. Use the Tools dropdown in the Server Manager window to navigate to Active Directory Users and Computers tool.

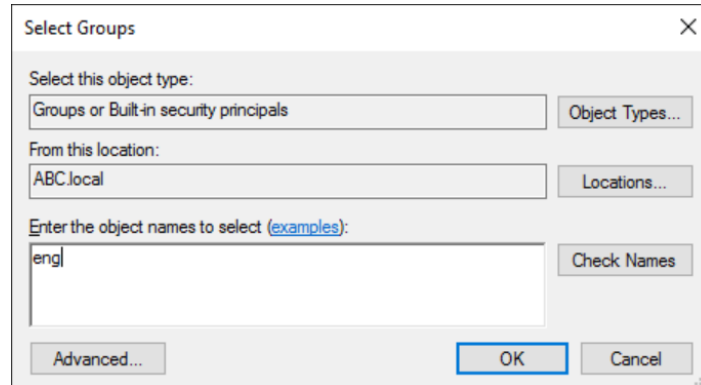
WINDOWS SERVER LAB



2. Click Staff dropdown, click department
3. While holding down control, select all of the users in department



4. Right click a selected user, click Add to a group...
5. Begin typing department, click Check names, click Ok, then click OK. Repeat steps 2 through 5 for each organizational unit



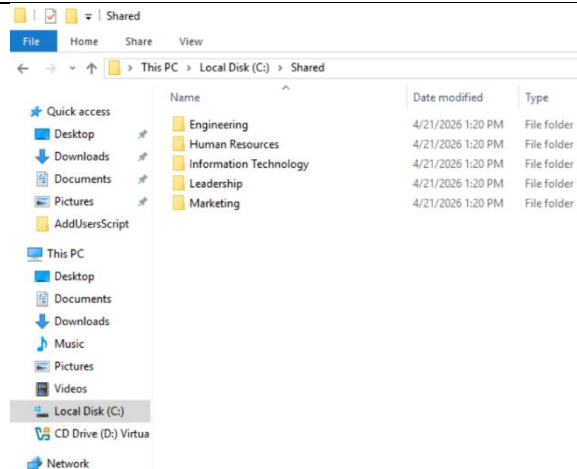
6. To add the Directors of each department to the Leadership Security Group, navigate to the following users:
 - Willard Smith
 - John Williams
 - Lupita Nyong'o
 - Samantha Rockwell
7. For each user, right click user, click "Add to group...", begin typing leadership, click Check names, Click Ok, then Click OK
8. To add the members of each department to the Cultural Committee Distribution Group, navigate to the following users:
 - Jackie McMullen
 - Janet Villanueva
 - Steven Ballmery
 - Lupita Nyong'o
 - Stephen Goss
9. For each user, right click user, click Add to group..., begin typing cultural, click Check names, Click Ok, Click OK
10. To provide IT Department Members with Domain Level Administrator rights, navigate to Active Directory Users and Computers tool
11. Click the Staff dropdown, click Information Technology, right click the Information Technology Security Group
12. Click "Add to group...", begin typing Domain Admins, click Check names, Click Ok, then Click OK

CREATING NETWORK SHARES

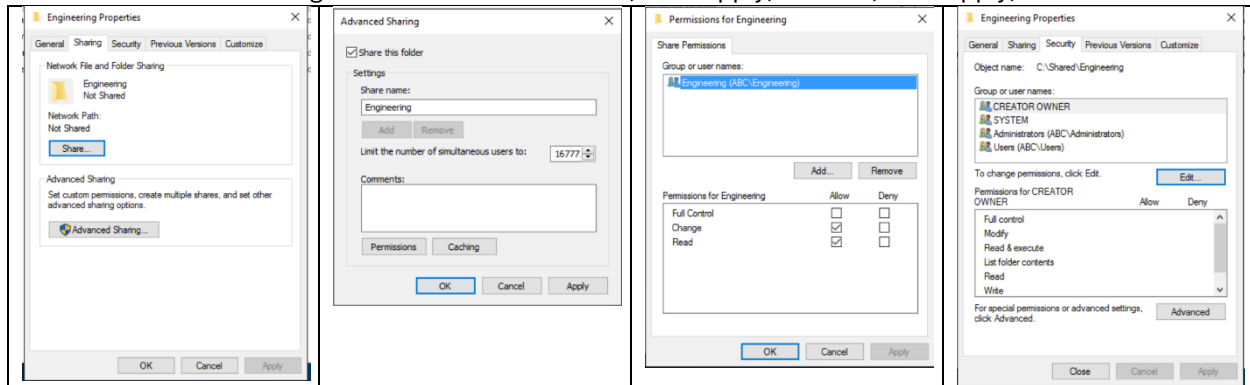
1. Open the file explorer, navigate to local disk(C:), right click empty space in the window for the C: drive, hover over New, then click Folder.
2. Type "Shared", press enter
3. Right click empty space in the window for the Shared folder, hover over New, then click Folder
4. Type department name, then press enter.
5. Repeat steps 3 & 4 for the following departments

WINDOWS SERVER LAB

- Engineering
- Human Resources
- Information Technology
- Leadership
- Marketing

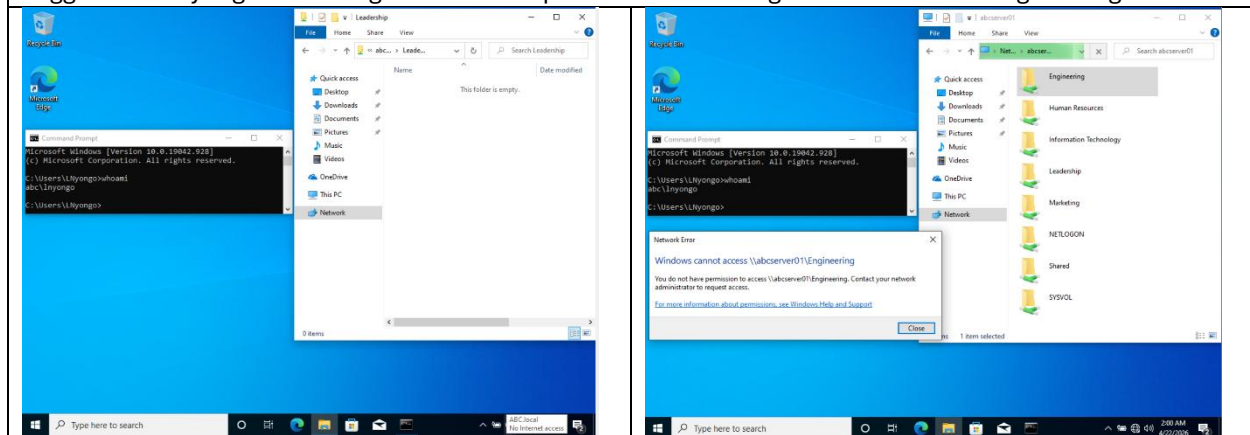


6. Right click a department folder, click properties.
7. Navigate to the Sharing tab, click advanced sharing.
8. Check share this folder, click Permissions
9. Click Remove, click add, enter the corresponding department name, click Check Names, click OK, then check “Change” and “Read” under Allow, click Apply, click Ok, click Apply, click Ok



10. Navigate to Security Tab in folder properties, click edit, click Add, enter the corresponding department name, click Check Names, click OK, check “Read”, “Write”, and “modify”, click Apply, click OK, then click Close. Repeat steps 6 through 10 for department folder in the shared folder.

Logged in as lnyongo accessing the Leadership folder and showing denied access to Engineering folder

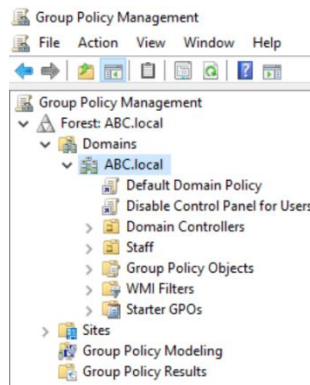


CREATE GPOS

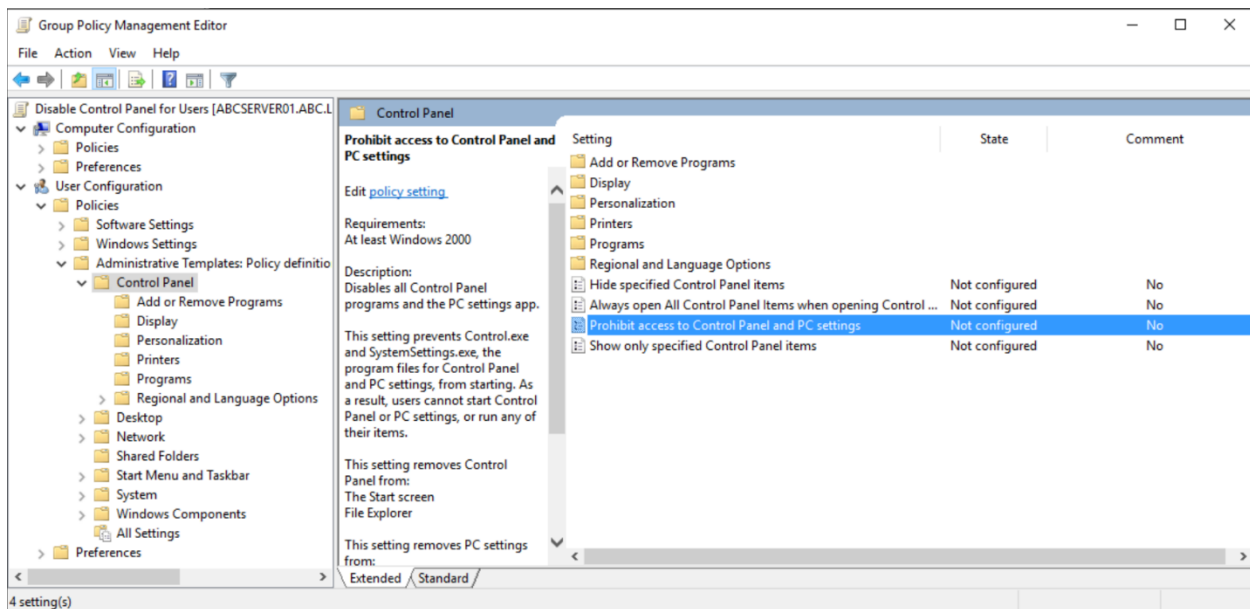
1. Use the Tools dropdown in the Server Manager window to navigate to Group Policy Management

WINDOWS SERVER LAB

- Use the dropdown for “Forest: ABC.local”, dropdown Domains, right click “ABC.local”, then click Create a GPO in this domain, and Link it here. A “New GPO” window will appear where you will name the GPO: Disable Control Panel for Users.

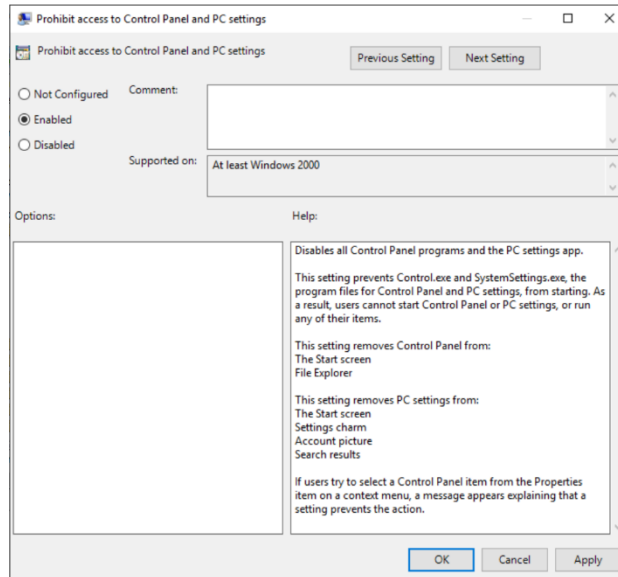


- Right click the new GPO, click edit. The Group Policy Management Editor window will appear.

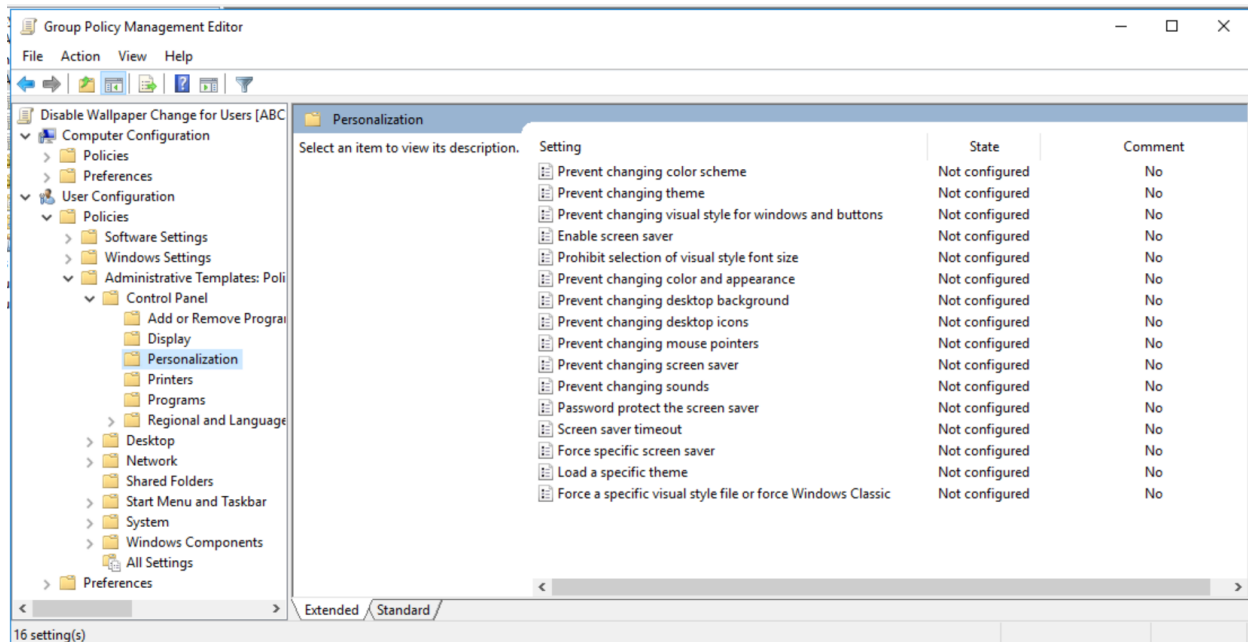


- Under User Configuration, dropdown Policies, dropdown Administrative Templates, click Control Panel, then double click “Prohibit access to Control Panel and PC settings”. A Prohibit access to Control Panel and PC settings window will appear

WINDOWS SERVER LAB



- Set to Enabled, click Apply, Click OK, close Group Policy Manager Editor Window
- Right click "ABC.local" in the Group Policy Management window, click "Create a GPO in this domain, and Link it here...", type "Disable Wallpaper Change for Users", then click ok
- Right click the new GPO, click edit, Under User Configuration dropdown Policies, dropdown Administrative Templates, dropdown Control Panel, click Personalization, then double-click "Prevent changing desktop background"



- Set to enabled, click Apply, click OK, then close Group Policy Manager Editor Window.

WINDOWS SERVER LAB

The screenshot shows the 'Prevent changing desktop background' Group Policy setting window. The window has a title bar with the text 'Prevent changing desktop background' and standard window controls. Below the title bar, there is a tabbed interface with 'Previous Setting' and 'Next Setting' tabs. The 'Previous Setting' tab is selected. The main area contains three radio buttons: 'Not Configured', 'Enabled' (which is selected), and 'Disabled'. To the right of these buttons is a 'Comment:' text box. Below the radio buttons is a 'Supported on:' dropdown menu showing 'At least Windows 2000'. At the bottom left, there is an 'Options:' section with a large empty box. At the bottom right, there is a 'Help:' section with a text area containing the following text: 'Prevents users from adding or changing the background design of the desktop. By default, users can use the Desktop Background page in the Personalization or Display Control Panel to add a background design (wallpaper) to their desktop. If you enable this setting, none of the Desktop Background settings can be changed by the user. To specify wallpaper for a group, use the "Desktop Wallpaper" setting. Note: You must also enable the "Desktop Wallpaper" setting to prevent users from changing the desktop wallpaper. Refer to KB article: Q327998 for more information. Also, see the "Allow only bitmapped wallpaper" setting.' At the bottom right of the window are 'OK', 'Cancel', and 'Apply' buttons.

Prevent changing desktop background

Previous Setting Next Setting

☐ Not Configured ☒ Enabled ☐ Disabled

Comment:

Supported on: At least Windows 2000

Options:

Help:

Prevents users from adding or changing the background design of the desktop.

By default, users can use the Desktop Background page in the Personalization or Display Control Panel to add a background design (wallpaper) to their desktop.

If you enable this setting, none of the Desktop Background settings can be changed by the user.

To specify wallpaper for a group, use the "Desktop Wallpaper" setting.

Note: You must also enable the "Desktop Wallpaper" setting to prevent users from changing the desktop wallpaper. Refer to KB article: Q327998 for more information.

Also, see the "Allow only bitmapped wallpaper" setting.

OK Cancel Apply